

RAVITEJA MADDELA

MSc Computer Science & Engineering Student | Ex-Software Developer (6+ Yrs Exp)

Copenhagen, Denmark +45 71806856 s252744@student.dtu.dk tejamaddela9@gmail.com github.com/teja-mr-1

linkedin.com/in/raviteja-m-33a413263

PROFESSIONAL SUMMARY

MSc Computer Science student at the Technical University of Denmark (DTU) with 6+ years of industry experience as a Software Developer across healthcare, fintech, and banking domains (GE Healthcare, Swiss Re, American Express). Specializing in Machine Learning systems, Multi-Agent pathfinding, web applications, and scalable backend infrastructure using PyTorch, Java, and Python.

EDUCATION

Technical University of Denmark (DTU) 2025 – Present

Master of Science in Computer Science & Engineering (Copenhagen, Denmark)

Key Focus: Machine Learning Systems, Multi-Agent Systems, Algorithm Engineering, Distributed Systems. Student ID: s252744.

RVR & JC College of Engineering (Acharya Nagarjuna University) 2012 – 2016

Bachelor of Technology in Information Technology (India)

TECHNICAL SKILLS

AI & ML Systems: PyTorch, QLoRA, Transformers, RAG, Mixture-of-Experts (MoE), BioClinicalBERT, FAISS
Backend & Web: Python, Java, JavaScript, Leaflet.js, Mapbox, FastAPI, Spring Boot, REST APIs, React, Tailwind CSS
Cloud & Tools: Docker, Git, Linux, AWS, Kaggle, HuggingFace, Netlify

FEATURED ENGINEERING & RESEARCH PROJECTS

Sparse MoE-QLoRA for Danish Legal QA PyTorch MoE QLoRA Personal Research Project

- Engineered a 4-bit Quantized Low-Rank Adaptation (QLoRA) pipeline to fine-tune a 7B LLM for Danish legal domain QA.
- Implemented centroid-based Sparse Mixture-of-Experts (MoE) routing using E5 embeddings paired with FAISS RAG retrieval.
- Achieved 6.88 tokens/sec generation throughput and sub-5ms in-place VRAM adapter hot-swapping on a single Kaggle T4 GPU.

Weather-Aware Route Planner & Travel Advisor JavaScript Leaflet.js Mapbox Personal Project

- Built an interactive route planning application combining Leaflet.js and Mapbox GL JS map rendering.
- Integrated OpenWeatherMap APIs to sample weather conditions along route waypoints in real time.
- Implemented Netlify proxy functions for secure API key handling, with planned integration of LLM AI API services for automated route weather verdicts.

Multi-Agent Coordination in Hospital Domain Java CBS A* Search DTU 02285 Course Project

- Developed a Java multi-agent path finding (MAPF) solver for autonomous robot agents in the Mavis Hospital Domain grid world.
- Implemented Hungarian matching for box-goal allocation, fast greedy priority orderings, and 3-phase box staging.
- Built a Conflict-Based Search (CBS) repair engine with CAT bias, opportunistic bypass, Lemma-1 jump points, and segment replanning.

Multi-Modal Clinical Phenotyping System PyTorch FastAPI FHIR R4 Healthcare AI Project

- Built a dual-encoder PyTorch pipeline pairing BioClinicalBERT for unstructured clinical notes with a custom Masked Lab-Test Transformer (MLTM) for numerical lab features on MIMIC-IV data (3,397 patient records, 25 conditions, 0.88 AUROC).
- Developed a FastAPI inference service with SHAP lab importance, BERT attention scores, FHIR R4 bundle adapters, and Gradio web interface.

PROFESSIONAL EXPERIENCE

Software Developer Jan 2023 – Dec 2023

Infinite Computer Solutions (Client: GE Healthcare) (Bengaluru, India)

- Engineered internal license management software for hospital equipment (MRI and CT scanners), enabling staff to easily activate, update, and manage feature licenses without manual intervention.
- Streamlined automated installation and upgrade tooling, reducing system downtime and manual overhead for active hospital scanners.

Software Developer

Feb 2019 – Jul 2022

Tech Mahindra (Client: Swiss Re) (Hyderabad, India / Zurich, Switzerland)

- Developed *SwiftMX*, a transaction conversion tool mapping insurance business team requests into automated bond and security purchase orders via third-party financial tools (Apex).
- Built testing and data migration utilities for TRAX application upgrades, verifying config transfer success rates to guarantee zero downtime.
- Implemented custom encryption middleware routing TRAX payment files (salary and insurance disbursements) through multi-system networks (Swift) securely to banking institutions.

Junior Software Engineer

Sep 2016 – Jan 2019

Tata Consultancy Services (Client: American Express) (Hyderabad, India)

- Developed online PDF credit statement generation features for American Express Mexico internet banking, replacing inaccessible XML formats.

CERTIFICATIONS & LANGUAGES

Certifications: Python Institute Certified Programmer | Lean Six Sigma Yellow Belt | TOEFL iBT (Score: 94/120)

Languages: English (Fluent), Telugu (Native), Danish (Prøve i Dansk 2 / PD2)